



Graphic Era
HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Santhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

GEHU/02E/9.1.3

End Term (Odd) Semester Examination ^{NOV} 2025

Roll no. 2492001

Name of the Program and semester: BCA 5

Name of the Course: Introduction to Dot Net Programming

Course Code: TBC505

Time: 3-hour

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question are 20 (twenty).
- (iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

- a. Differentiate between .NET Framework and .NET Core in terms of architecture and performance. CO1
- b. Compare value types and reference types in C# with appropriate code examples. CO2
- c. Assess the benefits of using StringBuilder instead of the String class in C#. CO2

Q2. (2X10=20 Marks)

- a. Illustrate the process of compilation and execution in .NET using a suitable diagram. CO1
- b. Write a C# program to demonstrate the use of Jagged Arrays to store and display student marks. CO2
- c. Differentiate between Connected and Disconnected architectures in ADO.NET with suitable examples. CO5

Q3. (2X10=20 Marks)

- a. Explain the concepts of Boxing and Unboxing in C#. Demonstrate with a C# program. CO2
- b. Write a C# program to perform CRUD operations using SqlConnection and SqlCommand class CO5
- c. What is the purpose of Exception Handling in C#? Illustrate with an example. CO4

Q4. (2X10=20 Marks)

- a. Explain the use of abstract classes and interfaces in object-oriented programming with the help of examples. CO3
- b. Describe the process of creating and using a DLL (Dynamic Link Library) in C#. How can it be referenced in another project? CO4
- c. Create a custom exception class named InvalidAgeException and throw it when the user enters age below 18. CO4

Q5. (2X10=20 Marks)

- a. Explain the architecture of ADO.NET with a labeled diagram. CO5
- b. Write a C# program that demonstrates constructor overloading and method overloading. CO3
- c. Write a C# program to count vowels in a string. CO2